

RUSSELLVILLE, AR, USA

WMO: 723429

Lat: 35.258N

Lon: 93.095W

Elev: 403

StdP: 14.48

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 53920

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.2	21.5	4.6	7.1	26.3	9.0	8.9	27.6	20.4	45.2	18.4	42.6	2.6	320	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	98.9	76.2	95.6	76.6	93.1	75.9	79.9	91.4	78.8	90.2	77.9	89.0	5.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.0	142.5	84.7	75.7	136.6	83.7	74.9	132.6	82.9	43.6	91.2	42.5	90.3	41.5	88.9	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.6	13.5	11.4		10.5	101.3	4.6	4.8	7.2	104.8	4.5	107.6	2.0	110.3	-1.3	113.9	
				9.4	81.9	4.2	1.2	6.3	82.8	3.9	83.5	1.5	84.2	-1.5	85.1	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.6	40.1	43.2	52.9	61.4	69.8	78.3	81.4	81.0	73.9	62.2	51.2	42.7	(6)
		DBStd	16.63	9.69	9.75	9.35	7.73	6.76	4.61	4.46	5.06	6.27	8.38	8.89	8.86	(7)
		HDD50	1013	334	235	76	9	1	0	0	0	7	89	262		(8)
		HDD65	3237	773	612	388	155	36	0	0	7	155	420	691		(9)
		CDD50	5255	28	44	166	352	614	850	973	960	717	387	127	37	(10)
		CDD65	2002	1	1	13	47	185	400	508	496	274	69	7	1	(11)
		CDH74	20762	2	12	129	483	1596	4143	5630	5344	2692	665	65	1	(12)
		CDH80	9514	0	0	15	96	505	1920	2885	2738	1157	191	7	0	(13)
(14)	Wind	WSAvg	3.7	4.7	4.9	5.1	4.7	3.6	2.8	2.5	2.6	2.9	3.3	3.7	4.2	(14)
(15)	Precipitation	PrecAvg	49.8	3.7	3.1	4.6	5.0	5.4	3.8	3.6	3.0	3.9	4.1	4.8	4.4	(15)
(16)		PrecMax	68.1	7.3	8.1	13.9	11.6	12.7	8.4	6.6	7.0	9.1	10.7	12.2	10.2	(16)
(17)		PrecMin	35.9	0.7	0.9	1.6	1.7	0.6	1.3	1.2	0.6	0.6	0.3	1.0	0.2	(17)
(18)		PrecStd	8.8	1.9	1.7	2.7	2.4	2.8	1.6	1.4	1.5	2.0	2.5	3.2	2.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	72.1	75.5	82.1	86.4	91.1	99.5	103.5	104.1	97.3	89.1	81.0	72.1		(19)
(20)		MCWB	61.6	62.3	64.4	68.8	72.3	74.0	74.6	76.5	73.9	70.8	67.5	63.9		(20)
(21)		DB	67.7	70.2	78.0	82.7	88.0	95.0	98.7	99.2	93.1	85.3	74.9	67.5		(21)
(22)		MCWB	59.2	58.8	63.0	66.2	71.8	76.4	76.7	76.4	73.7	70.6	63.2	60.1		(22)
(23)		DB	62.9	65.7	73.3	79.6	85.5	92.2	95.3	95.6	90.4	81.3	71.3	63.0		(23)
(24)		MCWB	56.3	56.4	60.9	65.1	71.3	75.6	77.4	76.4	73.4	67.5	61.6	56.4		(24)
(25)		DB	57.2	61.2	69.7	76.1	82.5	89.7	92.5	92.6	87.1	77.2	67.2	58.0		(25)
(26)		MCWB	49.0	53.3	58.6	63.7	70.4	74.9	76.8	76.1	72.1	66.0	59.0	52.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	64.9	65.2	67.7	72.2	76.5	80.2	81.4	81.2	78.0	75.2	69.6	65.6		(27)
(28)		MCDB	69.3	71.6	77.8	82.0	85.6	91.4	91.9	93.4	89.3	84.5	76.5	70.4		(28)
(29)		WB	62.2	62.4	65.4	69.7	74.9	78.6	80.1	79.8	76.7	73.1	66.5	62.1		(29)
(30)		MCDB	66.6	67.9	75.0	78.1	83.5	90.0	91.0	91.9	87.6	80.9	72.6	66.6		(30)
(31)		WB	56.7	58.7	63.1	68.2	73.4	77.5	79.0	78.7	75.5	70.9	63.7	57.7		(31)
(32)		MCDB	60.7	64.2	70.8	76.1	81.9	88.2	90.0	90.4	85.3	77.5	68.6	61.4		(32)
(33)		WB	50.2	53.6	60.5	66.2	72.1	76.3	78.0	77.7	74.3	68.7	60.3	53.6		(33)
(34)		MCDB	56.1	59.6	67.5	73.5	80.0	86.2	88.8	89.1	83.3	74.6	66.2	57.1		(34)
(35)	Mean Daily Temperature Range	MDBR	21.3	21.1	23.4	23.2	20.7	20.6	20.9	21.1	23.2	23.6	23.0	20.0		(35)
(36)		5% DB	28.9	29.1	27.9	27.4	23.4	23.6	24.4	25.4	25.9	27.0	27.4	25.8		(36)
(37)		MCWBR	19.5	18.6	15.4	13.5	10.0	8.3	7.1	7.6	9.0	12.8	17.1	18.6		(37)
(38)		5% WB	21.3	22.4	23.0	21.0	19.4	20.3	20.6	21.5	21.4	21.2	22.4	20.9		(38)
(39)		MCWBR	17.5	16.9	14.4	12.2	9.8	8.3	7.7	7.7	8.4	11.2	16.5	17.3		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.321	0.353	0.390	0.421	0.452	0.478	0.475	0.402	0.355	0.332	0.308		(40)
(41)		taud	2.493	2.489	2.428	2.329	2.303	2.231	2.204	2.188	2.378	2.487	2.506	2.542		(41)
(42)		Ebn at Noon	280	293	291	284	275	265	257	256	272	277	273	279		(42)
(43)		Edh at Noon	27	30	34	40	42	45	46	45	36	30	26	24		(43)
(44)	All-Sky Solar Radiation	RadAvg	788	941	1295	1625	1772	2011	1964	1814	1554	1186	847	686		(44)
(45)		RadStd	69	121	90	103	155	175	122	128	122	156	96	81		(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page